

J-Anne Yow

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Summary

Final-year PhD researcher in Robotics at NTU Singapore focused on building adaptive robotic systems that learn from and respond to user interaction. Experienced in integrating reinforcement learning, control strategies, and foundation models into end-to-end robotic systems. Passionate about designing generalizable, user-informed robots, and eager to further apply foundation models in policy learning to build more adaptive and scalable robotic systems.

Education

Nanyang Technological University, Singapore

Sept 2021 – Aug 2025

Ph.D. in Mechanical Engineering

(Expected)

- **Supervisor:** Prof Ang Wei Tech
- **Proposed Dissertation:** Enhancing Human-Robot Interaction for Personalized Robot Behaviour in Assistive Feeding
- **Research Interests:** Human-Robot Interaction, Machine Learning, Foundation Models, Continual Learning

Nanyang Technological University, Singapore

Aug 2017 – May 2021

BE in Mechanical Engineering with a Second Major in Business

- **GPA:** 4.92/5.00, Dean's List every Academic Year
- **Awards:** Dr Leung Shiu Kee Gold Medal Award, ASEAN Undergraduate Scholarship

Research Experience

Robot-Assisted Feeding

2022 – present

- Led the development of an end-to-end robot-assisted feeding system, including bite sequencing, adaptive food acquisition and safe bite transfer.
- Designed a personalized feeding pipeline that adapts to user preferences using verbal corrections, leveraging foundation models to interpret and respond to user feedback.
- Developed a goal-conditioned scooping policy using reinforcement learning to acquire targeted bite sizes across diverse food properties.
- Preparing the system for real-world deployment and user trials with elderly users to better understand user preferences and refine robot behavior for safe, effective assistive feeding.

Grasping in Clutter

2020 – 2023

- Developed a point-and-click interface for supervised pick-and-place in cluttered environments, integrating object segmentation and grasp cycling for improved grasp selection.
- Developed a POMDP-based shared autonomy framework that queries users under intent uncertainty, improving goal disambiguation, enabling more effective human-robot collaboration.

Work Experience

Research Assistant, Nanyang Technological University – Singapore

Aug 2021 – present

- Led the development of a personalized robot-assisted feeding system, integrating perception, control, and language-based user interaction for adaptive assistance.
- Managed project planning, resource coordination, and ethical and safety compliance for system development and user trials.
- Contributed to funding proposals and helped initiate research collaborations with overseas institutes.
- Scoped and mentored undergraduate final-year projects, providing guidance on research direction, methodology and technical implementation.

SaaS Sales Operations Intern, ByteDance – Singapore

May 2020 – July 2020

- Designed and optimized dashboards to support operational insights and data-driven decision-making across the Lark APAC team.

- Improved Salesforce data quality and streamlined workflows, enhancing consistency across regional business units.
- Analyzed user behaviour trends (e.g., daily active users (DAU), tenant health scores) to support go-to-market strategy and assess product-market fit.

Engineering Intern, Oishii – New Jersey, USA

Jan 2020 – Mar 2020

- Oversaw a project to prepare infrastructure for future system expansion, balancing technical requirements and operational constraints.
- Planned and implemented sensor-actuator systems to support automation across farm operations, including wiring, integration, and layout decisions.
- Collaborated with cross-functional contractors to resolve infrastructure limitations and improve farm operations.

Journal Publications

ExTraCT - Explainable Trajectory Corrections for language-based human-robot interaction using Textual feature descriptions

Sept 2024

J-Anne Yow, Neha P Garg, Manoj Ramanathan, Wei Tech Ang

Frontiers in Robotics and AI

Shared Autonomy of a Robotic Manipulator for Grasping under Human Intent Uncertainty using POMDPs

Nov 2023

J-Anne Yow, Neha P Garg, Wei Tech Ang

IEEE Transactions on Robotics (T-RO)

Other Publications

Simulating Safe Bite Transfer in Robot-Assisted Feeding with a Soft Head and Articulated Jaw

May 2025

Yi Heng San, Vasanth Ravichandram, *J-Anne Yow*, Sherwin Chan, Wei Tech Ang

International Conference on Rehabilitation Robotics (ICORR)

Design of a Breakaway Utensil Attachment for Enhanced Safety in Robot-Assisted Feeding

May 2025

Hau Wen Chang, *J-Anne Yow*, Wei Tech Ang

International Conference on Rehabilitation Robotics (ICORR)

Adaptive Scooping in Simulation for Assistive Feeding: Meeting User Preferences in Bite Size (Best Paper Award)

Oct 2024

J-Anne Yow, Neha P Garg, Wei Tech Ang

Workshop on Interactive Robots and AI for Healthcare, IEEE/RSJ International Conference on Intelligent Robots and Systems

Service

Reviewer: IEEE Robotics and Automation Letters (RAL), IEEE International Conference on Robotics and Automation (ICRA)

Skills

Programming: Python, C/C++, R, SQL

Robotics & Tools: ROS, MuJoCo, Motion Planning, Simulation, Robot Learning, Computer Vision, Linux

Languages: English, Chinese, Malay